

CaTSCAN: Calculating Typed Subgraph Counts Analytically for Networks

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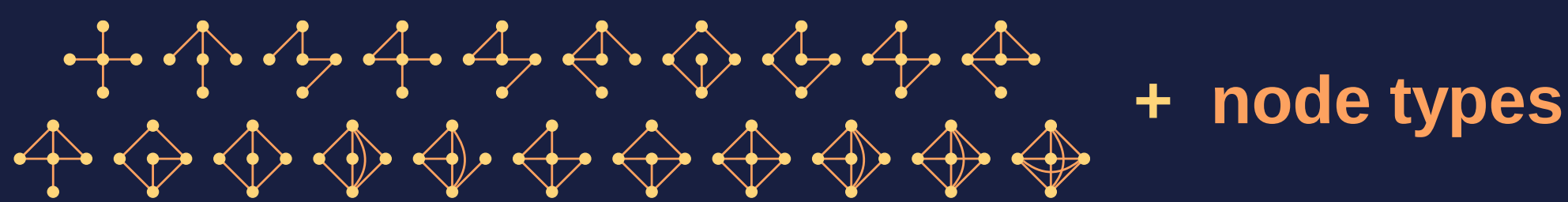


1 A CAT-scan for Typed Networks

- In a **Heterogeneous Information Network (HIN)** nodes have **types**
- What **typed building blocks** (= subgraphs) is an HIN made of?



2 The Gap: They can Only Count to Four



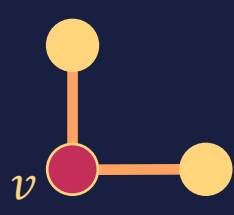
CaTSCAN is the **first** to count **typed** subgraphs with up to **5 nodes**.

3 The Ideas that Count

- **Graphs of interest are large:** millions of nodes and edges
- **Combinatorial explosion:** trillions (10^{12}), or even quintillions (10^{18}) of subgraph instances
- **Enumerating** individual instances **is bound to fail!**
→ avoid enumeration!

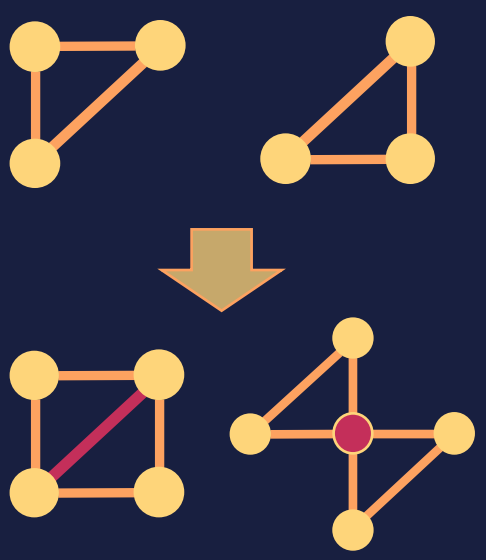
Don't count – calculate!

```
def count_typed_wedges():
    wedges = { }
    for v in V do:
        for t0 in T do:
            wedges[t_v, t0, t0] +=  $\binom{\deg_{t_0}(v)}{2}$ 
            for t1 in T, t0 < t1 do:
                wedges[t_v, t0, t1] +=  $\deg_{t_0}(v) \cdot \deg_{t_1}(v)$ 
    return wedges
```



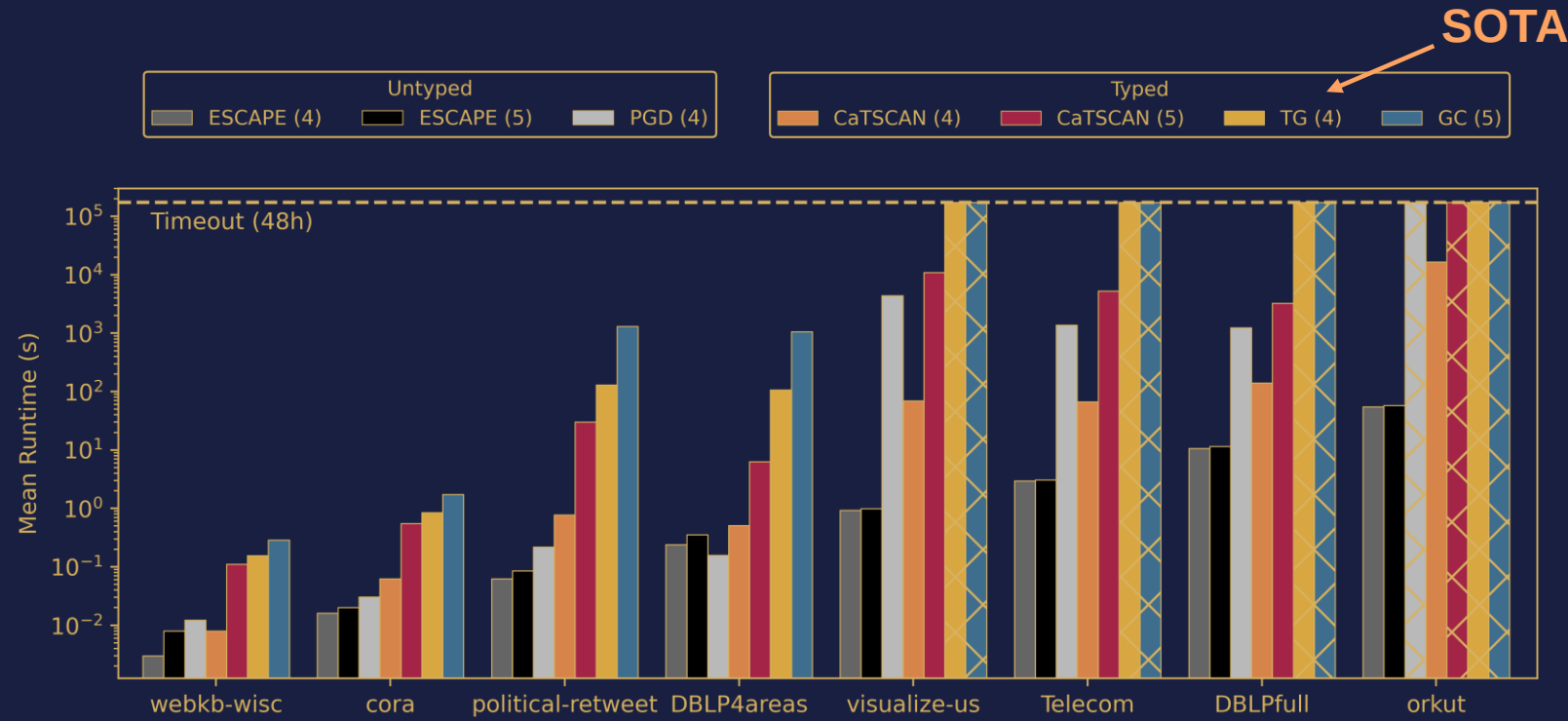
Combine counts

... of smaller subgraphs for larger subgraphs



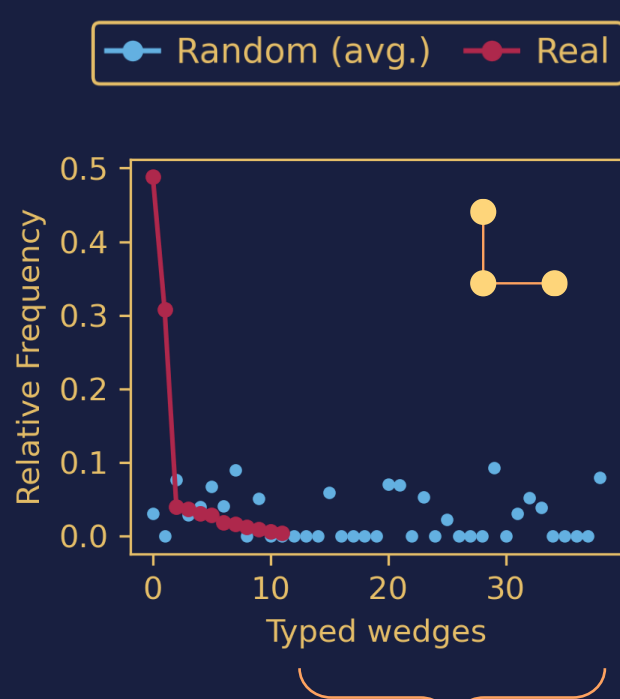
There are **28 more...** see our paper for details.

4 Only Untyped Counting is Faster...



- Counts **typed 4-node** subgraphs **1-4 OoM** faster than SOTA
- Counts **typed 5-node** subgraphs faster than **4-node** SOTA

5 ... but Typed Counts are More Informative



- DBLP4areas: shuffle node types (100 runs)
- Count frequencies of typed subgraph configurations
- Real frequencies are **systematic**, not random!

6 What Now?

Got ideas? Talk to me: →

- Consider types on **edges**, including **direction** and **multi-edges**.
- **Parallel** and distributed solution (work in progress)